

One-Month Plan for Jibril

1. Immediate comments on the work done so far

- Sections 5 and 6, as currently added, damage the paper structure because they duplicate material that should remain inside Section 4 (Experimental Results and Discussion).
- Table 1 and Table 12 do not necessarily indicate an error by themselves: Table 1 reports baseline validation results, whereas Table 12 reports the final tuned models on validation and test. However, the draft does not explain this transition clearly enough, so a reader may think the results are inconsistent.
- For stochastic or subset-based pipelines, repeated runs with fixed seeds must be documented. Otherwise, part of the difference between tables may come from run-to-run variability rather than only from tuning.
- The most serious scientific weakness is that some models are trained or tuned on subsets while others use much larger portions of the dataset. This makes the comparison unfair and weakens the conclusions.
- The large validation-to-test degradation remains the key technical issue. The next month must therefore focus on three goals simultaneously: fair comparison, stronger temporal evaluation, and transition from benchmarking to a first creative model idea that better matches Pergamon's deployment needs.

2. Writing and experiment rules you must follow from now on

1. Do not create new top-level paper sections without checking whether the content already belongs inside an existing section.
2. Every result table must state clearly whether it reports baseline results, tuned validation results, or final test results.
3. When a number changes between two tables, the text must explain why: tuning, different sample size, different metric, fixed-seed rerun, or stochastic variability.
4. Any comparative claim must be based on the same sample size for all models being compared. If a model cannot be trained on that agreed sample, it must be excluded from that comparison or presented in a separate computational-budget analysis.
5. Every new experiment must end with a saved artifact: a CSV table, a figure, a script, or a short memo.

3. Objective of this one-month plan

The goal of the next month is not to continue a simple comparative study. The comparison phase must now be converted into a controlled diagnostic phase that helps you understand which mechanisms work, why the current models degrade on future data, and how to design a new regression approach that better serves Pergamon Perceptive Technologies. A second objective is to address fairness by forcing all compared models to use the same sample size. To support both

fairness and future creativity, you must also begin exploring coreset selection as a principled way to build a representative reduced training set on which all candidate models can be compared.

4. One-month task plan

Week 1 – Build a fair benchmark protocol and explain the current inconsistencies

Task 1.1: Audit Table 1 versus Table 12

You must create a model-by-model note for the five selected models (Gradient Boosting, Spline Regression, Polynomial Regression, TabNet, and MLP) explaining why the validation values in Table 12 differ from those in Table 1. You must explicitly state which numbers are baseline numbers and which correspond to tuned models. For TabNet and any other stochastic pipeline, you must also report the random seed and whether repeated runs change the results.

Task 1.2: Define one common training budget for all compared models

You must choose one fixed sample size that all compared models can use fairly. This sample size must be feasible for the five models in Table 12 and also for the stronger additional candidates: XGBoost or LightGBM or CatBoost, plus Extra Trees or HistGradientBoosting. The chosen size must be justified in writing from both a computational and a scientific point of view.

Task 1.3: Build the first fair subset baseline

Using the chosen common sample size, you must construct a first reduced dataset for fair comparison. The reduced set must preserve temporal order and should keep the target distribution as representative as possible. You should compare at least two reduction mechanisms: chronological downsampling and stratified-by-target bin sampling. The output of this task is a fixed reduced dataset that can be reused by all models in later weeks.

Task 1.4: Short memo on fairness and current weaknesses

At the end of the week, you must write a concise memo answering three questions: why the current draft is scientifically weakened by unequal sample sizes, whether Table 1 versus Table 12 differences are fully explained by tuning, and what fixed-size benchmark protocol will be used from now on.

Expected deliverables for Week 1

- One note that reconciles Table 1 and Table 12 clearly.
- One fixed common-sample benchmark specification.
- One reusable reduced dataset for fair model comparison.
- One short fairness memo.

Week 2 – Compare all candidate models fairly on the same data budget

Task 2.1: Re-evaluate the core models on the same sample size

You must retrain and evaluate the five selected models from Table 12 on the same reduced dataset and under the same temporal split protocol. No model is allowed to use more data than the others in this main comparison.

Task 2.2: Add stronger baselines under the same fairness rule

You must add at least two stronger tree-boosting candidates from the following set: XGBoost, LightGBM, CatBoost, Extra Trees, or HistGradientBoosting. These models must be run on exactly the same reduced dataset and evaluated with the same metrics. This point is essential because some of these models may prove stronger than the current five under equal conditions.

Task 2.3: Use rolling temporal validation

You must replace the single validation split with at least three rolling temporal validation windows. In each window, the training block must occur earlier in time than the validation block. You must then report mean and standard deviation of RMSE, MAE, and R^2 across windows for every compared model.

Task 2.4: Add runtime and model-size analysis

For every candidate model, you must measure training time, inference time, and, if possible, saved-model size or approximate footprint. Pergamon will care about the trade-off between accuracy and practical deployment cost, not only raw validation performance.

Expected deliverables for Week 2

- One fair comparison table across all candidate models.
- One rolling-validation table with mean $\pm$ std.
- One runtime / model-size table.
- One short note stating which model is currently best under equal data budget.

Week 3 – Explore coreset selection and convert the benchmark into a creative direction

Task 3.1: Enter the coreset-selection direction

You must study coreset selection as a principled way to build a representative reduced training set instead of using arbitrary subsampling. You should begin with simple and implementable strategies such as random coreset, k-means prototype selection, farthest-point sampling, and density-aware sampling. The objective is to test whether a better reduced set improves fairness and model quality at the same budget.

Task 3.2: Compare coreset strategies with the fixed-size baseline

Using the same sample size chosen in Week 1, you must compare several coreset strategies on the regression task. Each strategy should be used to train the same leading models. You must then determine whether some models are especially sensitive to the quality of the selected subset.

Task 3.3: Propose a first creative model direction for Pergamon

Based on the comparative study and the coreset results, you must propose one first original direction for the project. This should not be another generic benchmark. Acceptable directions include a coreset-aware spline model, a spline-plus-boosting residual model trained on a representative coreset, a regime-aware model that separates low- and high-concentration behavior, or a teacher-student compression pipeline where a strong regressor teaches a lighter deployable model.

Task 3.4: Explain transferability to classification

Because XGBoost, LightGBM, CatBoost, Extra Trees, and HistGradientBoosting can also be used for classification, you must actively coordinate with Otmane when analyzing these model families. Otmane has already evaluated some closely related classifier baselines in his classification study, including XGBoost, Random Forest, and Gradient Boosting, so your work should build on that existing effort rather than proceed in isolation. For each regression model you analyze, you must add a short note identifying which mechanisms, behaviors, or design choices may transfer to the classification branch, and you should discuss these points with Otmane before finalizing your conclusions. You do not need to run classification experiments this week, but you must document the potential bridges between the regression and classification sides of the project and identify where collaboration can help both directions converge toward a more useful deployment-oriented solution for Pergamon.

Expected deliverables for Week 3

- A coreset-comparison table on the regression task.
- A recommendation on which subset-selection mechanism is most suitable.
- One written proposal for the first creative model direction.
- One short transferability note from regression to classification.

Week 4 – Prototype the new direction and fix the paper structure

Task 4.1: Implement the first prototype

You must implement one first prototype of the creative direction selected in Week 3. The model does not need to be final, but it must be executable and compared against the best fair baseline from Week 2. The comparison must again use the same sample size and rolling temporal validation protocol.

Task 4.2: Interpret the prototype scientifically

You must analyze whether the prototype helps because of the subset-selection mechanism, the nonlinear backbone, the residual correction, the regime split, or the compression idea. The goal is to identify which ingredient is promising, not only whether one run was lucky.

Task 4.3: Repair the paper structure

You must integrate the added material correctly into the paper. Section 4 should remain the main experimental section. Baseline comparison, fairness protocol, coreset analysis, and tuned-model

evaluation should appear as coherent subsections there rather than as separate top-level Sections 5 and 6.

Task 4.4: Write a two-part closing note

At the end of the month, you must write a short internal report with two parts: first, what was learned scientifically during the fair comparison and coreset study; second, what the next-month plan should be in order to transform the prototype into a deployable model for Pergamon.

Expected deliverables for Week 4

- One first prototype of the new model direction.
- One fair comparison of the prototype versus the best baseline.
- One cleaned paper structure plan.
- One end-of-month internal report with the next technical roadmap.

5. Main message you must understand

The comparative study is not the project's endpoint. It is a controlled diagnostic phase whose purpose is to reveal what kind of model structure fits the problem and what kind of reduced training set is fair and representative. From now on, your work must move toward a more creative and company-relevant contribution: a regression approach that is fair to compare, aware of data budget constraints, and capable of evolving toward deployment. The same model families and subset-selection ideas may later be useful for both regression and classification branches of the project.